

Ranking Candidates the Way a Great Recruiter Would

Redrob Hackathon — Intelligent Candidate Discovery & Ranking Challenge

Five-dimension rule-based ranker • no embeddings, no LLM calls, no GPU • 100K candidates in ~100s on CPU

The Trap We Were Explicitly Warned About

- The JD says it plainly: "the right answer is not candidates whose skills section contains the most AI keywords."
- Keyword stuffers: every buzzword in the skills array, irrelevant title (e.g. "Marketing Manager").
- Plain-language Tier 5s: never wrote "RAG" or "Pinecone", but built a recommendation system at a product company.
- Behavioral twins: identical-looking skill profiles, very different recruiter-response and activity signals.
- ~80 honeypots: internally impossible profiles forced to relevance tier 0 in ground truth.
- Conclusion: score career-history substance, not skills-array vocabulary.

Architecture: Five Independent, Inspectable Dimensions

Technical Depth

40%

Embeddings/retrieval, vector-DB, ranking systems, eval frameworks, Python — scanned from career text

Title Relevance

30%

Is the current title actually AI/ML/search-flavored, vs adjacent, vs unrelated

Logistics Fit

20%

Pune/Noida/Tier-1 location or relocation, notice period, 5-9 yr experience band

Disqualifier ×

0.25–1.0×

JD's explicit "don't wants": pure research, recent LangChain-only, architecture-without-code, job-hopping, consulting-only, CV/speech without NLP

Behavioral ×

0.7–1.0×

Activity recency, recruiter response rate, open-to-work, interview completion, offer acceptance — a modifier, not a core driver

$$final_score = (0.40 \cdot technical + 0.30 \cdot title + 0.20 \cdot logistics) \times disqualifier_mult \times (0.7 + 0.3 \cdot behavioral)$$

Why No Embeddings, No LLM Calls, No GPU

- Compute budget: 5 minutes, 16GB RAM, CPU-only, zero network during ranking, for 100,000 candidates.
- An LLM call per candidate cannot fit that budget — the spec says this outright, and frames it as the real-world constraint.
- Regex/keyword scanning over career-history text is $O(n)$ and fully local: ~100 seconds for the full pool on a laptop CPU.
- Every score is traceable to a one-line reason built from the same signals — nothing hallucinated for the reasoning column.
- Trade-off, acknowledged: regex misses paraphrases the lexicon doesn't anticipate. Chosen deliberately over an embedding model that would violate the no-GPU/no-network rule.

Honeypot Handling: Internal-Consistency Checks, Not Special-Casing

- "Expert" proficiency in a skill used for ~0 months.
- Multiple concurrent "is_current" roles, or end_date before start_date.
- Total career_history tenure that doesn't plausibly match stated years_of_experience.
- Career span far shorter than the claimed years of experience.
- Flagged profiles are forced to a near-zero score, not deleted from the pool — they still appear, they just sink.
- Result on the full 100K-candidate pool: 52 honeypots detected overall, 0 in the top 100.

The JD's "We Explicitly Do Not Want" Section, Operationalized

JD signal	How it's detected	Penalty
Pure research, no production deployment	industry mentions research/academia AND no production-deployment language anywhere in career text	×0.25
Recent LangChain/OpenAI-only AI experience	LangChain/OpenAI skill <12mo AND no pre-LLM ML signal (TF-IDF, word2vec, XGBoost, IR, etc.)	×0.50
Architecture/lead title, no recent coding	title matches architect/lead/manager AND 18+ months in role with no production signal in that role's description	×0.55
Title-chasing	3+ jobs, 2+ with rising seniority words, average tenure <18 months	×0.60
Consulting-only career	every employer in career_history is a pure services/consulting firm	×0.45
CV/speech/robotics, no NLP/IR	CV/speech/robotics keywords present, NLP/IR keywords absent	×0.50

Results on the Full 100,000-Candidate Pool

- Runtime: ~102 seconds end-to-end on a CPU-only laptop — well inside the 5-minute budget.
- Honeypots: 52 detected in the full pool; 0 surfaced in the top 100 (limit is 10%).
- Top-100 spot check: titles are AI/ML/search-flavored (Search Engineer, Staff ML Engineer, Lead AI Engineer, Applied Scientist), at recognizable product companies, spanning the 5–9 year band with sensible falloff outside it.
- Submission CSV passes the official `validate_submission.py` with zero errors: 100 rows, unique ranks 1–100, non-increasing scores, valid `candidate_ids`.
- Reasoning column is per-candidate and grounded only in fields actually present on that profile — no templated name-insertion, no hallucinated skills.

Honest Limitations

- No labeled ground truth is available, so dimension weights are reasoned from the JD text and spot-checked on output — not fit by optimization. A learning-to-rank layer is the natural next step if feedback data becomes available.
- Regex lexicons are finite; an unusual phrase for "vector database" that isn't in the keyword list will be missed. This is the deliberate cost of staying CPU-only/offline rather than using an embedding model.
- Disqualifier penalties are multiplicative heuristics, not hard rules — a strong candidate can still rank highly despite one weak signal, which mirrors how the JD itself says it weighs trade-offs ("we'll seriously consider candidates outside the band if other signals are strong").